Business Requirements Document

Solar | MES Pre-Sales Discovery | 02 May 2026

1. Executive Summary

MES implementation for a **Solar** manufacturer producing **solar panels** at **300 MW annually**. Current systems: manual tracking in spreadsheets.

2. Business Objectives & Pain Points

• lack of production tracking and scheduling

• no quality traceability

• inadequate machine integration

• absence of real-time OEE dashboards

3. Functional Requirements

#	Requirement
1	production tracking and scheduling
2	quality traceability (batch genealogy from wafer to module)
3	OPC UA machine integration
4	real-time OEE dashboards
5	

4. Non-Functional Requirements

Category	Requirement
	99.5% uptime during production shifts
Performance	Real■time data refresh ≤ 5 seconds
	Role■based access control; audit logs retained 3 years
Scalability	Architecture must support 2× production line growth
	RTO ≤ 4 hours; RPO ≤ 1 hour
Integration	ERP: SAP via REST/IDoc; OPC UA for machines

5. Success Criteria

✓ Production tracking accuracy ±2%

✓ Full batch genealogy from wafer to module

✓ Real■time quality alerts ≤ 10 seconds

✓ ERP–MES synchronisation latency ≤ 30 seconds

✓ OEE dashboard available within 1 hour of go■live

6. Key Stakeholders

Role	Responsibility	Involvement
	Executive sponsor	Sign■off
IT Manager	Infrastructure & security	Architecture review
	Shop■floor champion	UAT lead
Quality Manager	QMS requirements	Requirements & UAT
	ERP integration point	Integration testing

7. Assumptions & Constraints

• Customer will provide network access to all production machines.

• SAP system landscape is stable and not under major upgrade during implementation.

• At least one super■user per production line nominated before Phase 2.

• Hardware procurement (servers, HMIs, scanners) is the customer's responsibility.